Section 2.3 — Factoring (Basic Methods)

Math 103: College Algebra

1 Factoring Numbers

List all factors of each number.

1. 21
2. 36
3. 24
4. 147

Give the prime factorization of each number.

5. 35
6. 28
7. 36
8. 42

2 Factored Form

Identify whether each polynomial is written in a factored form (that is, as a product of simpler polynomials).

9. $(x+4)(x-3)$
10. $2x(x+5)-3y^2(x-4)$
11. $7x^2y^5z^{12}$
12. $\left(x^2-4\right)(x+3)-32$
13. $18x^2y^3\left(2y^2-5x^5\right)$
14. $\left(x^2-2x+4\right)^3(x+5)$

3 Greatest Common Factor

Find the greatest common factor of each list of monomials. You may find it helpful to factor the coefficients into primes.

15. $4x^2yz^2$, $6xy^5z$
16. $28x^2$, $49y^5$
17. $18r^2s^3t^5$, $42s^5t^5v^5$, $36r^4s^2t^{11}$
18. $32xy^2$, $24x^2y$, $72y^2z$, $8x^2y^2$

4 Factoring the GCF

For each polynomial, factor out the greatest common factor. You may find it helpful to factor the coefficients into primes.

19. x^2-x
20. $18r^4-27r^6$
21. $w^3v^2+w^2v+wv^2$
22. $21t^4u^3-105t^5u^3v+63t^6u^2v^4$
23. $10g+4g^3-100g^4+120g^5$
24. $4a^3y^3-6ax^2y^2+4axy^2$

For each polynomial, factor out a **negative** GCF.

25. $-30b^2-24b$
26. $-15x^4y+21x^2y^2$
27. $-6x^2-17x+24$
28. $-21a^4bc^3-15a^3b^3c^3+27a^2b^2c^4$

5 Factoring by Grouping

Factor each polynomial by grouping.

29. $cx+cy+bx+by$
30. $xy-xw-yz+wz$
31. $y^4+4y^3-by-4b$
32. $1+x+x^2+x^3$
33. x^3-4x^2+x-4
34. $12+3y-4x-xy$
35. $a^3m-a^3n-bm+bn$
36. $z^3-4z^2-4z+16$